

$$\begin{bmatrix} 3 & 0 & -3 & 1 & 2 \end{bmatrix} \quad w = \begin{bmatrix} 1 & 0 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & -1 \end{bmatrix} \rightarrow 3 + 0 + 3 = 6$$

$$\begin{bmatrix} 1 & 0 & -1 \end{bmatrix} \rightarrow 1 \cdot 0 + 0 \cdot (-3) + 1 \cdot (-1) = -1$$

$$\rightarrow 1 \cdot (-3) + 0 \cdot 1 + -1 \cdot (2) = -5$$

$$\begin{bmatrix} 6 & -1 & -5 \end{bmatrix}$$

1D-Convolution Operation

$$x = [x_1 \ x_2 \ \dots \ x_n] \quad w = [w_1 \ w_2 \ \dots \ w_k] \\ (\text{filter/kernel})$$

$$x \otimes w = S \quad \text{where} \quad S_i = \sum_{j=1}^k x_{i+j-1} w_j$$

$$S_1 = x_1 w_1 + x_2 w_2 + \dots + x_k w_k$$

$\vdots$

$$S_{n-k+1} = x_{n-k+1} w_1 + \dots + x_n w_k$$

$$S_{[n-k+1]}$$

$\Rightarrow$  2D - Convolution

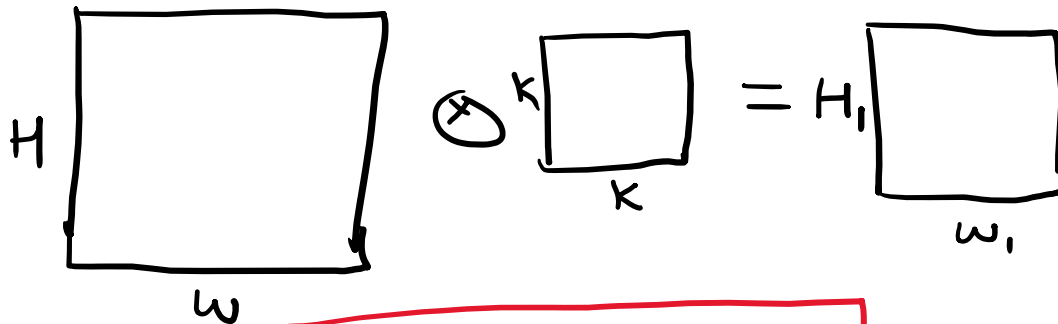
$$\begin{array}{|c|c|c|c|c|} \hline \circ & \circ & \circ & \circ & \circ \\ \hline \circ & 3 & 2 & 1 & 4 & 5 \\ \hline \circ & 0 & 1 & 2 & 0 & 1 \\ \hline \circ & 0 & 1 & 3 & -1 & 1 \\ \hline \circ & 2 & -1 & 0 & 1 & 0 \\ \hline \circ & \circ & \circ & \circ & \circ & \circ \\ \hline \end{array} \otimes \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 9 & 3 & 3 \\ 4 & 4 & 2 \end{bmatrix}$$

$f \quad 3 \times 3$   $2 \times 3$

# Padding

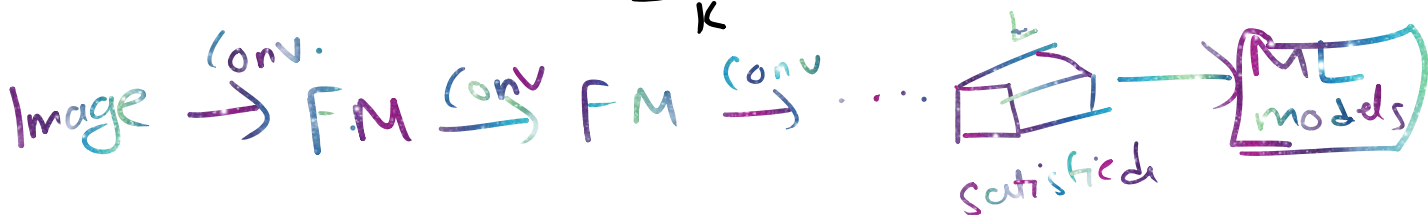
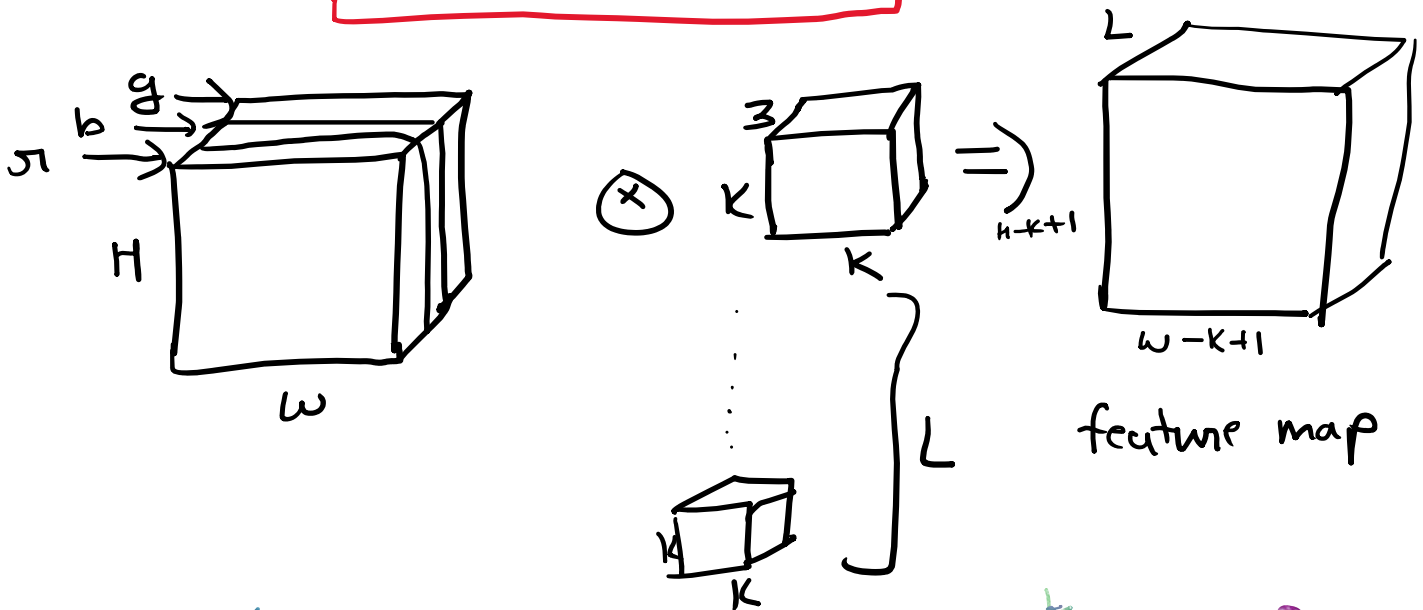
stride = 2

Padding = P, stride = S



$$W_1 = \frac{W - k + 1 + 2P}{S}$$

$$H_1 = \frac{H - k + 1 + 2P}{S}$$



## 2D conv examples

18 May 2026 17:32



$$\begin{matrix} * & \begin{matrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{matrix} & = \end{matrix}$$



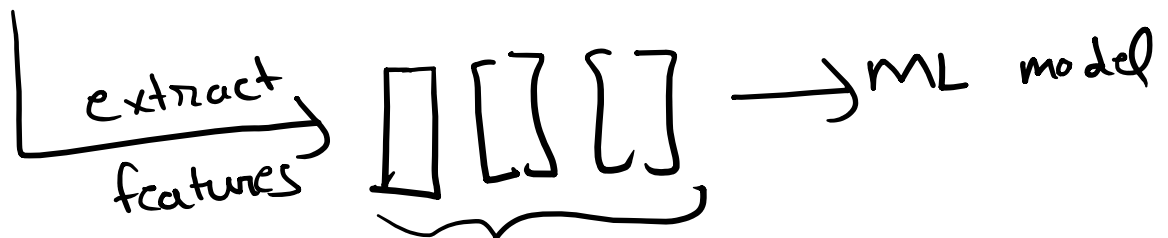
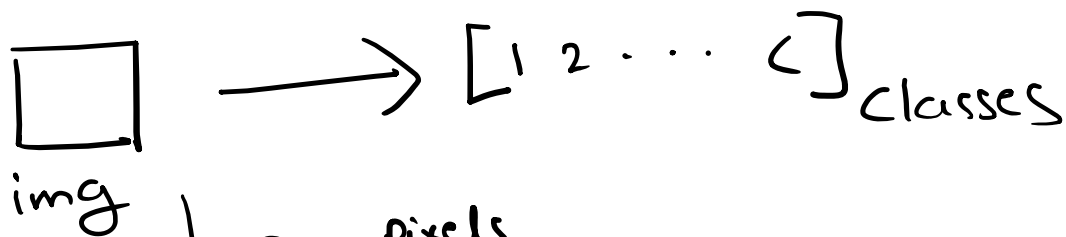
blurs the image



$$\begin{matrix} * & \begin{matrix} 1 & 1 & 1 \\ 1 & -8 & 1 \\ 1 & 1 & 1 \end{matrix} & = \end{matrix}$$



detects the edges

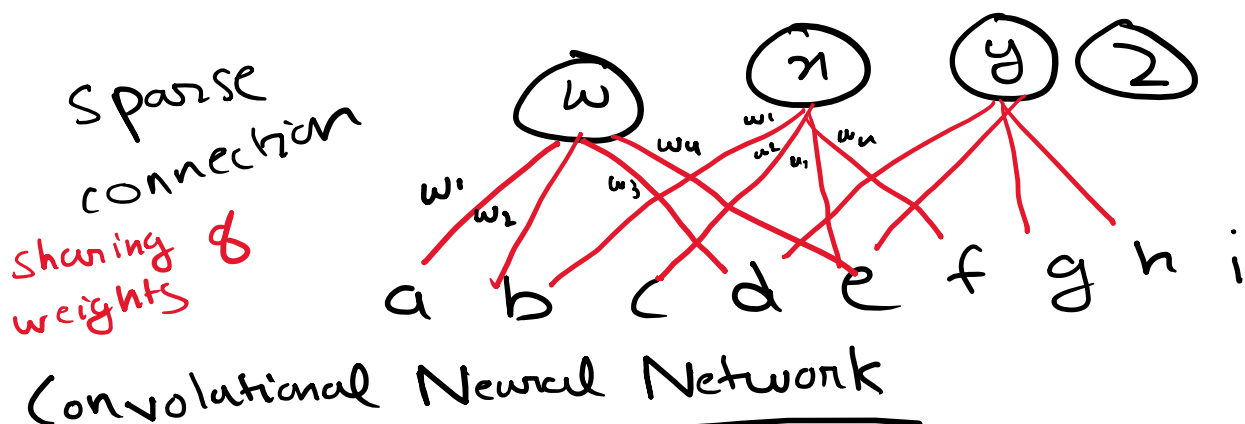
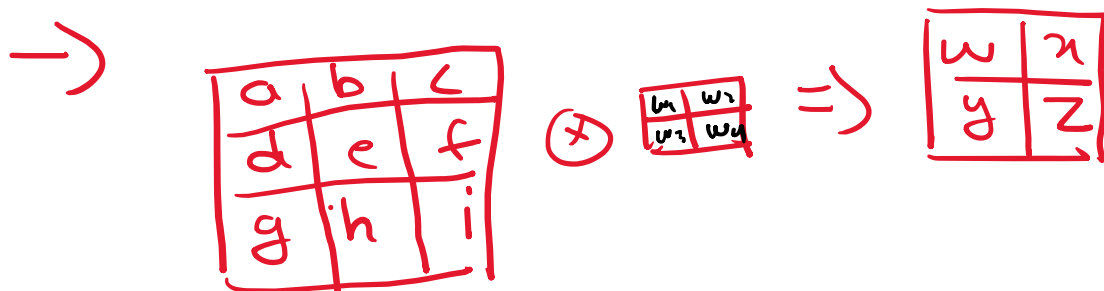


$\rightarrow$  using kernels

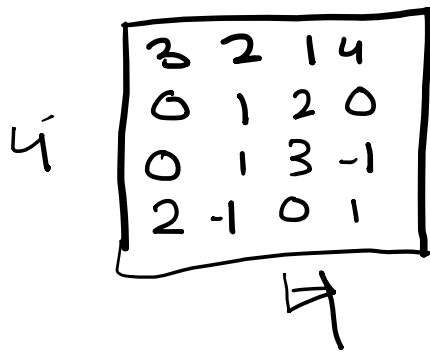
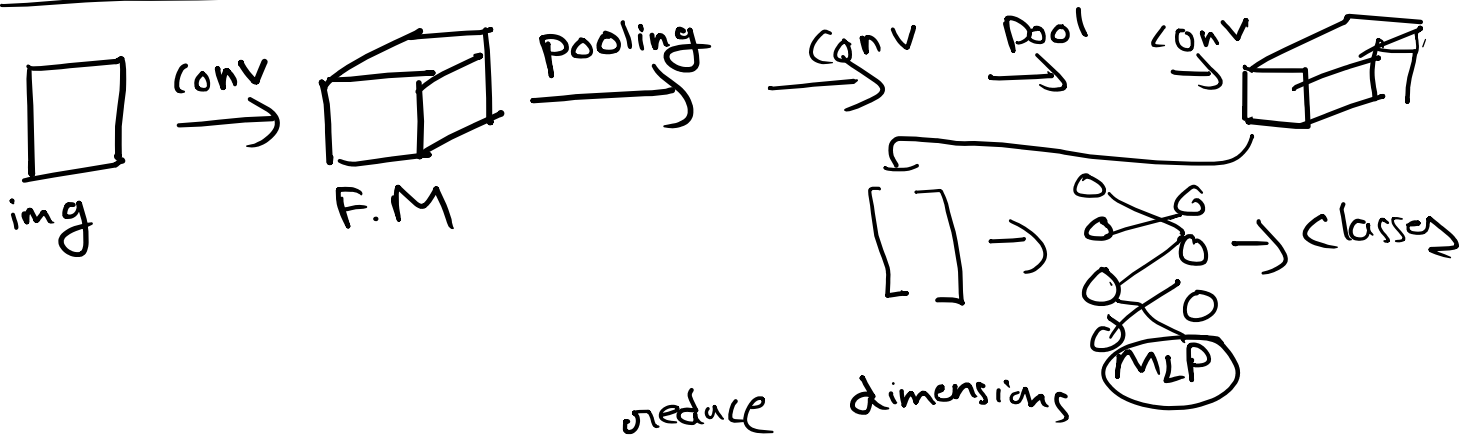
$\rightarrow$  SIFT / HOG

(static features)

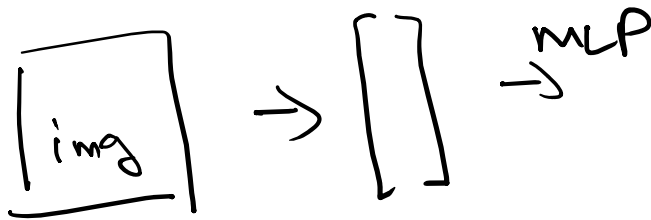
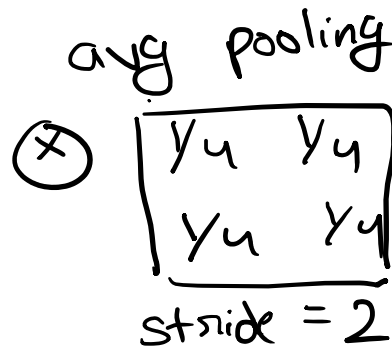
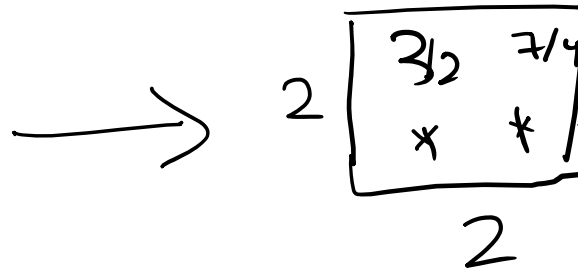
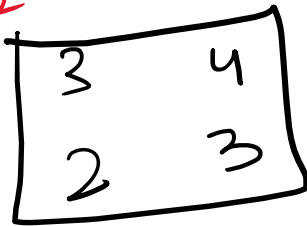
we can learn our kernel itself  
along with the classifier

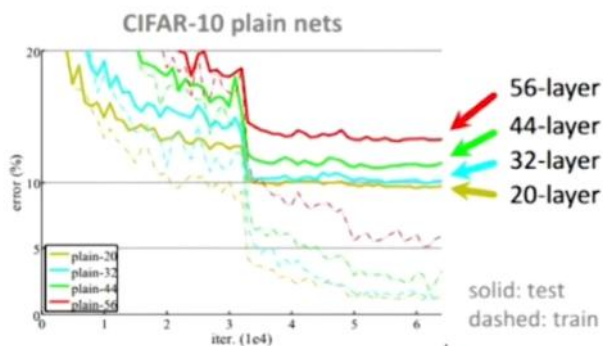


# Convolutional Neural Network

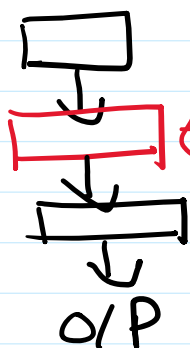
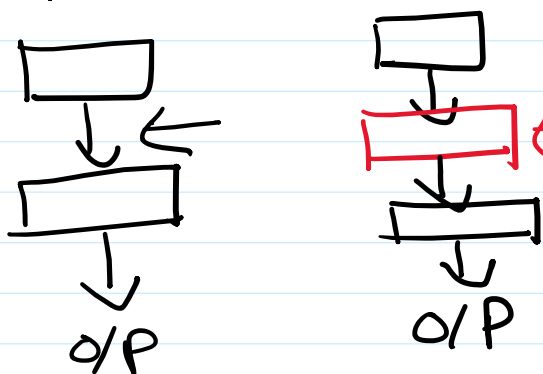


max pool  
2x2





Shallow network



learn identity mapping

2010 Imagenet

1000

Alexnet, GoogLeNet, Resnet

Hyperparameters

- ↳ filter size
- ↳ number of filters
- ↳ stride / padding
- ↳ # of layers

20-30

50  
100  
3.5%

$$y = F(x) + x$$

